

The EMT Simulation of the Study

What it is, how it was done, what it shows, and how to defend it

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In one sentence: The EMT run is a movie of one fault: it shows the real current wave during two fast reclosing shots and proves that fuse F671-2 survives the fast shots without DG (45 % of its melting heat) but melts during the second shot with DG (100 % at 0.752 s), because the DG raises the fuse current and keeps feeding the fault while the recloser is open.

1. What EMT is, in plain words

The rest of the study uses **short-circuit calculations**. They give one number per fault: the RMS current in steady state. That is a *photograph*. An **electromagnetic-transient (EMT) simulation** solves the network equations step by step in time (here every 0.1 ms) and gives the *instantaneous* current wave, including what happens when breakers open and close. That is a *movie*.

Question	Short-circuit (RMS) study	EMT simulation
What is calculated	one RMS current per fault	the current wave, point by point in time
Time	none: the fault is “frozen”	0 to 2 s in steps of 0.1 ms
DC offset, first peak	not shown	shown
Generator	one fixed reactance	the machine’s flux and rotor move: the current decays and the rotor can lose step
Breaker opening and re-closing	not possible	scripted as switching events
Used in the study for	all 39 cells, all settings, the classification	one fault, to check the reclosing <i>sequence</i>
Cost	seconds for hundreds of faults	one fault per run, three runs per case

Why the study needs it. The classification asks one question per fault: *does the fuse start to melt before the recloser’s fast trip?* It cannot say what happens over a whole reclosing sequence: trip, dead time, reclose, trip again. The fuse does not forget the first shot when the second one comes. Only a time-domain run can add the heating of successive shots and show what the DG does while the recloser is open.

2. The case that was simulated

Item	Value and reason
Fault	line-to-line, phases a–c, at node 684, through $0.2\ \Omega$ — the case of the reference method’s time-domain figure; node 684 only has phases a and c
Fault instant	0.30 s; the fault is permanent
Simulation	PowerFactory EMT (instantaneous values), step 0.1 ms, 0 to 2.0 s
Two runs	DG out of service, and DG in service (4.05 MVA synchronous machine at 692)
Settings	conventional (single-setting) R2: fast plug 300 A, TMS 0.1; delayed plug 600 A, TMS 1.0
Fuse watched	F671-2, A055C 300E, at the head of the lateral 671–684
Reclosing sequence	two fast shots, dead time 0.2 s (read from the reference figure), then the delayed curve
Breaker time	3 cycles = 0.050 s added to every recloser trip
Measured	current into node 632 (the plotted wave), through R2, and through the fuse; phases a and c

Where the devices are. The fault at 684 is *downstream* of R2. The fuse F671-2 is between node 671 and the fault. The DG at 692 is connected next to node 671, so it is *between R2 and the fuse*. This position is the whole story (Figure 1).

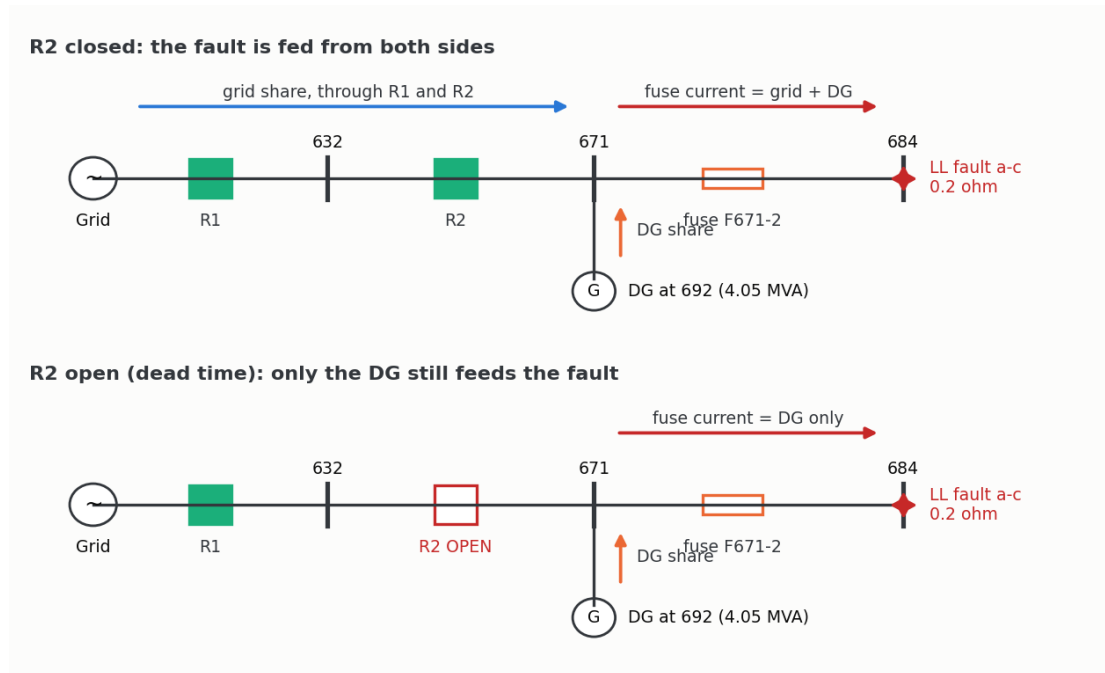


Figure 1. Who feeds the fault. With R2 open the grid is cut off, but the DG is still connected to the fault through the fuse.

3. How the simulation was run

PowerFactory's own relay and fuse elements were switched out of service for the EMT runs. Instead, every switching time was calculated from the *same curves used in the rest of the study*, applied to the simulated current. This keeps the sequence transparent: every time in the result can be traced to a curve and a current. It takes three passes per case.

Pass	What is simulated	What is taken from it
1	the fault only, no switching	the one-cycle RMS current through R2 → R2's fast time from its curve, plus the breaker time
2	fault + two fast trips and two recloses	the RMS current through the fuse at every time step → fuse heating; when it melts and when it clears; R2's delayed timer after the last reclose
3	the whole sequence, with the fuse opening	the final wave that is plotted in the report

3.1 From the wave to an RMS value

A relay curve and a fuse curve are given in RMS amperes, but the EMT gives instantaneous values. So the wave is converted with a **one-cycle moving RMS**: at every instant, the RMS of the last 1/60 s. The larger of phases a and c is used.

3.2 The recloser

$$t_{\text{trip}} = t_{R2, \text{fast}}(I_{R2}) + 3 \text{ cycles}$$

Without DG: $I_{R2} = 2607 \text{ A} \rightarrow$ curve time $0.070 \text{ s} + 0.050 \text{ s} = 0.120 \text{ s}$, so R2 opens at $0.30 + 0.120 = 0.420 \text{ s}$.
 With DG: $I_{R2} = 2436 \text{ A} \rightarrow 0.076 \text{ s} + 0.050 \text{ s} = 0.126 \text{ s}$, R2 opens at 0.426 s . R2 recloses 0.2 s later and the same fast time runs again.

3.3 The fuse: heat accumulation

A fuse melts when its element has absorbed enough heat. At a constant current I that takes the minimum-melting time $t_{MMT}(I)$ of its curve. When the current changes or is interrupted, each small time step Δt uses up a fraction $\Delta t/t_{MMT}(I)$ of the melting heat:

$$H(t) = \sum \frac{\Delta t}{t_{MMT}(I_{fuse}(t))}, \quad \text{the fuse melts when } H = 1 \text{ (100\%)}.$$

The same sum with the total-clearing time t_{TCT} gives the instant at which the fuse has cleared. **No cooling is assumed** while the recloser is open, which is on the safe side and reasonable for a dead time of 0.2 s.

4. Reading the current wave

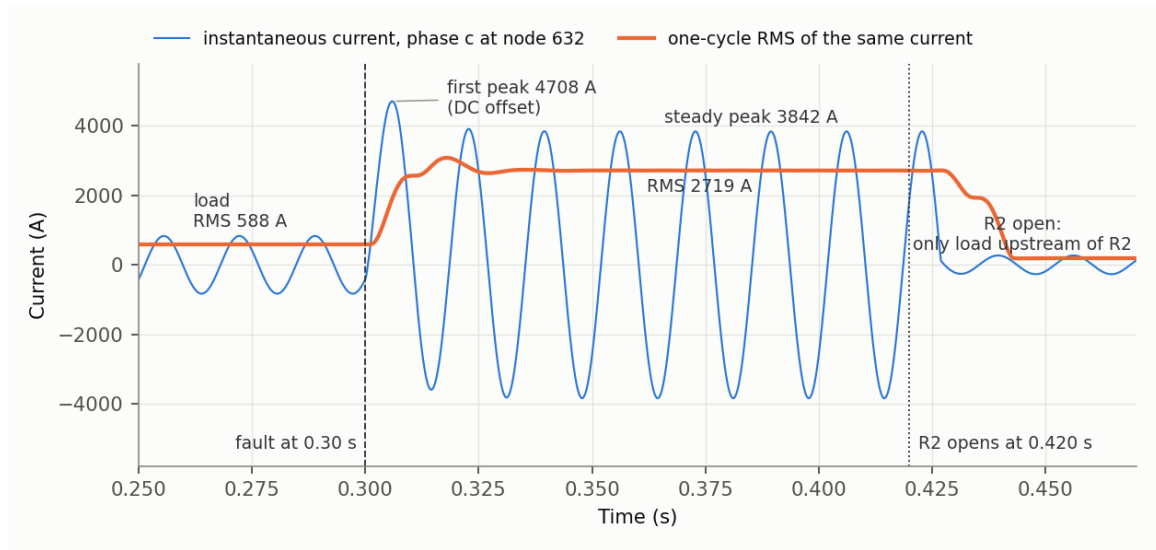


Figure 2. The first fast shot without DG, phase c of the current into node 632.

- **Before 0.30 s:** load current, 588 A RMS.
- **First peak 4708 A:** higher than the later peaks. This is the **DC offset**: the current in an inductive circuit cannot jump, so a decaying DC component is added to the AC wave. Its size depends on the instant of the fault on the voltage wave and on the X/R ratio; it is gone after about one cycle here.
- **Steady peak 3842 A, RMS 2719 A:** the ratio is $3842/2719 = 1.413 = \sqrt{2}$, as it must be for a sine wave. A short-circuit calculation only gives this RMS value.
- **0.420 s:** R2 opens. The current does not fall to zero at node 632, because the loads between the substation and R2 are still supplied (about 190 A).
- **The RMS line** needs one cycle to rise and one to fall: that is the one-cycle window.

	Without DG		With DG	
Current into node 632 (A)	phase a	phase c	phase a	phase c
Before the fault, RMS	557	588	261	286
During the fault, RMS	2362	2719	2443	2353
First peak	4288	4708	3805	3653
Steady peak	3338	3842	3517	3414
While R2 is open, RMS	97	191	91	185

With the DG the pre-fault current from the grid is about half (261 against 557 A): the DG supplies part of the load locally.

5. The result: the two sequences

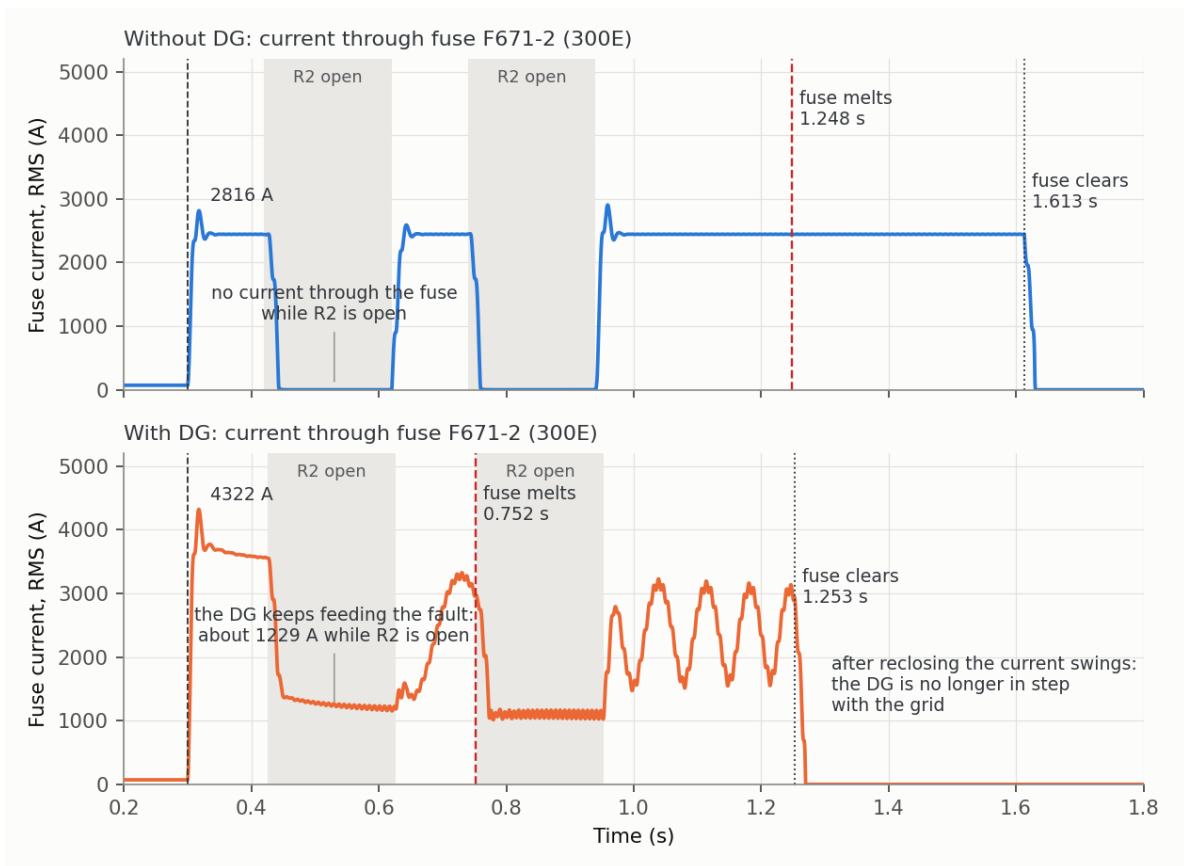


Figure 3. RMS current through fuse F671-2 over the whole sequence. Grey: R2 is open.

Event (time in s)	Without DG	With DG	Comment
Fault initiated	0.300	0.300	
R2 fast trip 1	0.420	0.426	R2 sees less current with DG
R2 reclose 1	0.620	0.626	dead time 0.2 s
Fuse F671-2 melts (with DG)	—	0.752	during fast shot 2
R2 fast trip 2	0.740	0.753	
R2 reclose 2	0.940	0.953	now on the delayed curve
Fuse F671-2 melts (without DG)	1.248	—	after both fast shots
Fuse F671-2 has cleared	1.613	1.253	
Fuse heat when the fast shots are over	45 %	105 %	of the melting heat

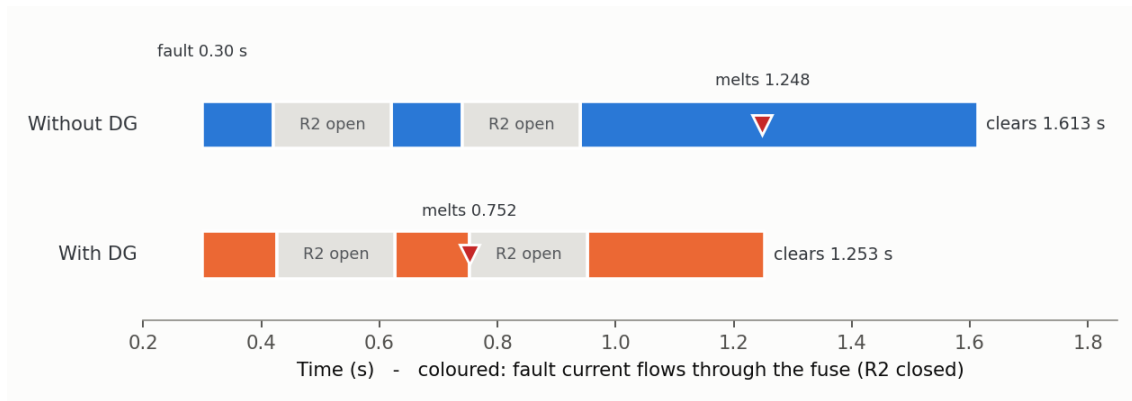


Figure 4. The two sequences on one time axis.

Without DG — fuse saving works. The fuse reaches only 45 % of its melting heat during the two fast shots. If the fault had been temporary, it would have been cleared by one of the two fast trips and the fuse would still be intact. Because the fault is permanent, it is still there after the second reclose; R2 is now on its delayed curve (2.67 s at this current), so the fuse has time to melt (1.248 s) and clear (1.613 s) *before* R2 would trip again. Only the lateral 671–684 is lost. This is exactly the intended sequence.

With DG — fuse saving is lost. The fuse melts at 0.752 s, one millisecond before the second fast trip. A temporary fault would now cost a fuse and a permanent outage of the lateral, which is what the fast shots were supposed to prevent.

6. Why the fuse melts with DG: the heat budget

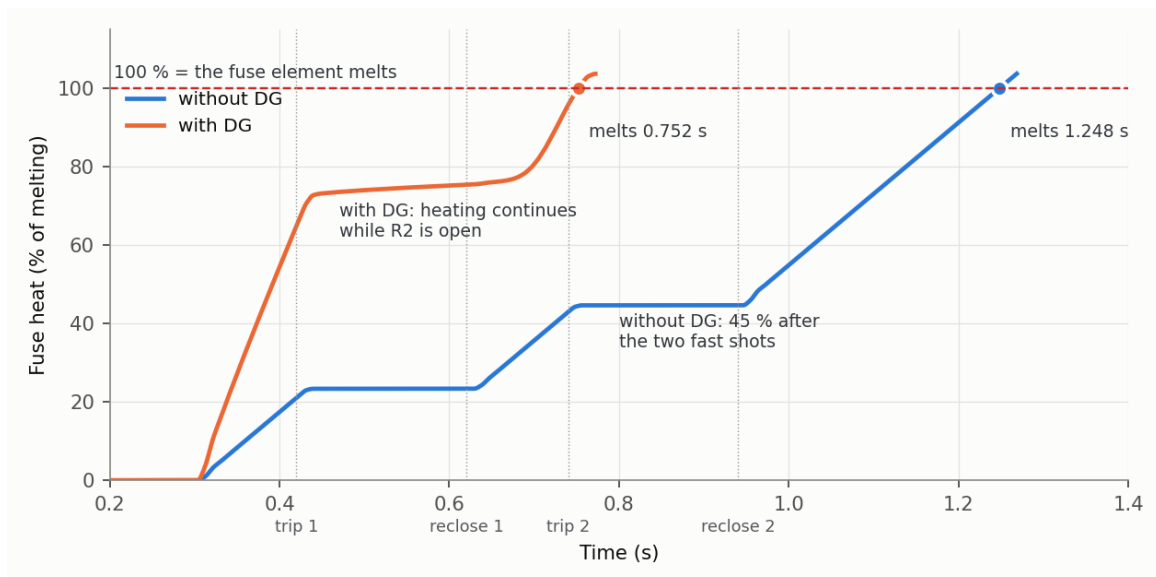


Figure 5. Accumulated fuse heat $H(t)$. Flat parts: no heating. The orange line keeps rising between trip 1 and reclose 1, because the DG still feeds the fault.

Interval	Without DG		With DG	
	heat added	total	heat added	total
Fast shot 1 (fault to trip 1)	21 %	21 %	68 %	68 %
Dead time 1 (R2 open)	2 %	23 %	7 %	75 %
Fast shot 2 (reclose 1 to trip 2)	20 %	43 %	25 %	100 %
Dead time 2 (R2 open)	2 %	45 %	5 %	105 %

The small amounts in the dead time without DG are the last cycle of fault current before the breaker contacts part (the RMS window is one cycle long).

Three things act together, all caused by the DG:

1. **The fuse carries more current.** 4322 A at the start against 2816 A without DG (grid + DG). At 4322 A the fuse melts in 0.120 s; at 2816 A in 0.361 s. One shot now costs 68 % instead of 21 %.
2. **The recloser sees less and is slower.** R2 carries only the grid share: 2436 A against 2607 A, so its fast shot lasts 0.126 s instead of 0.120 s.
3. **The DG keeps feeding the fault while R2 is open.** About 1230 A flows through the fuse during the dead time, adding 7 % and then 5 %. Without DG the dead time is a real pause.

A hand check you can do on the board

Without DG the fuse carries a steady 2441 A. Its curve gives $t_{MMT}(2441 \text{ A}) = 0.553 \text{ s}$. The fuse actually carried fault current for

$$0.120 \text{ (shot 1)} + 0.120 \text{ (shot 2)} + (1.248 - 0.940) \text{ (after reclose 2)} = 0.548 \text{ s}$$

before it melted. The 5 ms difference is the higher current of the first cycle of each shot (DC offset). So the simulated melting instant agrees with the fuse curve.

7. What the EMT showed that the curves alone could not

1. **Heating adds up over the shots.** The classification compares one fast trip with the melting time. For this fault with DG it would say the fuse survives one shot (68 %). The EMT shows that two shots plus the dead-time infeed do not fit into the fuse.
2. **The DG current decays.** The fuse current falls from 4322 A to about 3570 A within the first shot: a synchronous machine starts on its subtransient reactance and its contribution then falls towards the transient value. A short-circuit calculation gives only the initial value.
3. **The DG feeds the fault during the dead time.** This is invisible in any calculation that looks at one network state at a time. It also means the arc of a *temporary* fault may not extinguish while R2 is open — the purpose of the dead time is defeated.
4. **The DG falls out of step.** After each reclose the current through the fuse is no longer steady: it swings with a period of about 0.07 s (a difference of about 14 Hz between the DG and the grid). During the fault and the dead time the machine cannot deliver its power, so its rotor speeds up (its mechanical starting time is only 1.5 s), and R2 recloses onto a machine that is no longer synchronised.
5. **The margin matters.** At the initial 4322 A the fuse curve gives 0.120 s, and R2 with its breaker needs 0.126 s. With the zero-margin rule (curve time 0.076 s) the first shot looks safe; with the three-cycle breaker time it is at the edge. This is the same effect that reduces the held cells from 38 to 31 in the margin study.

8. How it fits the conclusions of the report

- It is a **time-domain confirmation** of the main cause of the problem: the fuse carries the grid and the DG current, the recloser only the grid share.
- It shows the **limit** stated in the report: for faults *downstream* of R2 the DG feeds the fault directly. The dual setting cannot help there, because R2's forward group is the same in both schemes and no setting of R2 changes what the DG delivers. That is why the EMT was run with the conventional settings.
- It supports the **future work** listed in the report: disconnection of the DG during the reclosing dead time, and a practical margin.

9. Limitations — say these yourself before you are asked

- **One fault only.** One location, one fault type, one fault impedance, conventional settings. It is a check of the sequence, not a second classification.
- **The protection is scripted.** Trip and melt instants come from the device curves applied to the simulated RMS current, in three passes; the relay and fuse elements of PowerFactory do not act in the EMT run. (They were checked separately: 1896 operating times within 1.22 %.)
- **The fuse is a heat sum,** without cooling and without an arc model. The clearing instant is the t_{TCT} sum, not a simulated arc.
- **The dead time (0.2 s) and the breaker time (3 cycles) are assumed;** the dead time was read from the reference figure.
- **The DG stays connected.** No anti-islanding or under/over-frequency protection is modelled. In practice the DG protection would be expected to trip it during the dead time, which would remove the dead-time infeed and the out-of-step reclosing — but not the higher fuse current of the first shot.
- **Step 0.1 ms** is right for the 60 Hz current and its DC offset; the run was not made to study switching over-voltages or travelling waves.
- **No DSDR case in EMT,** for the reason given in Section 8.

10. Questions you may be asked

Basics

Q1. What is an EMT simulation?

A simulation that solves the network's differential equations step by step in time and gives the instantaneous voltages and currents. Here: step 0.1 ms, 2 s, in PowerFactory.

Q2. How is it different from your short-circuit study?

The short-circuit study gives one RMS value for a frozen network. The EMT gives the wave: DC offset, decay of the generator current, and the effect of opening and reclosing. I used the short-circuit study for all 39 cells and the EMT for one fault, to check the sequence in time.

Q3. Why did you need it at all? The classification already tells you whether coordination holds.

The classification checks one fast trip against the fuse melting time. A reclosing sequence has two fast shots and dead times, and the fuse heating adds up. Only a time-domain run shows that, and shows what the DG does while the recloser is open.

Q4. Why this fault: LL a–c at 684 through 0.2 Ω ?

It is the fault of the reference method's time-domain figure, so the sequence can be set against it. Node 684 has only phases a and c, so an a–c fault is the largest phase fault there.

Q5. Why a step of 0.1 ms?

One 60 Hz cycle is 16.7 ms, so there are 167 points per cycle: more than enough for the fundamental current and its DC offset, which is all the relay and fuse curves need. I did not study switching overvoltages, which would need a smaller step and frequency-dependent line models.

The modelling

Q6. How is the recloser modelled?

As a three-phase switch on line 632–671 with scripted events. The trip instant is R2's fast-curve time at the simulated RMS current plus three cycles of breaker time; it recloses after a dead time of 0.2 s; two fast shots; after the second reclose the delayed curve applies.

Q7. Why did you switch off PowerFactory's own relay and fuse models?

To keep one consistent set of curves for the whole study and to make every instant traceable to a curve and a current. If both the scripted events and the built-in models acted, the sequence would be ambiguous. The built-in models were checked separately and agree within 1.22 %.

Q8. How do you get an RMS value out of an instantaneous wave?

A one-cycle moving RMS: at each instant the RMS of the previous 1/60 s. I take the larger of phases a and c.

Q9. How is the fuse modelled?

As accumulated heat: every time step adds $\Delta t/t_{MMT}(I)$. The fuse melts when the sum reaches 1. The same sum with t_{TCT} gives the clearing instant. It is the usual way to apply a time–current curve to a current that is not constant.

Q10. A fuse cools down during the dead time. Did you include that?

No. Cooling is neglected, which is on the safe side: it overestimates the heat slightly. With a dead time of 0.2 s the cooling of the element is small. Including it would not change the conclusion: with DG the fuse is at 68 % after one shot and the DG keeps heating it during the dead time.

Q11. Where do the 0.2 s dead time and the two fast shots come from?

From the reference figure: its reclosing instants are 0.2 s after the trips, and it shows two fast operations. Two fast and then delayed operations is also the usual fuse-saving sequence.

The results**Q12. Explain the 45 %.**

Without DG the fuse carries about 2441 A, where it needs 0.553 s to melt. Each fast shot lasts 0.120 s, so each uses $0.120/0.553 \approx 21\text{--}22\%$. Two shots: about 45 %. The fuse is intact after the fast shots.

Q13. Why does the fuse melt with the DG?

Three reasons. The fuse carries grid plus DG current, 4322 A instead of 2816 A, so it heats about three times faster. R2 carries only the grid share, 2436 A instead of 2607 A, so it trips slightly later. And the DG keeps about 1230 A flowing through the fuse while R2 is open. Total after two shots: 105 %.

Q14. The first peak is 4708 A but the later peaks are 3842 A. Why?

DC offset. The current cannot jump in an inductive circuit, so a decaying DC component appears; it depends on the fault instant on the voltage wave and on X/R. It is gone after about a cycle. The steady peak over the RMS is exactly $\sqrt{2}$.

Q15. With the DG the fuse current falls during the first shot. Why?

That is the synchronous machine: its fault contribution starts at the subtransient level and decays towards the transient level. A short-circuit calculation only gives the initial value; the EMT shows the decay.

Q16. After reclosing with the DG, the current swings up and down. What is that?

The DG has lost synchronism. During the fault and the dead time it cannot deliver its power, so the rotor accelerates; the machine is small, with a mechanical starting time of 1.5 s. When R2 recloses, the grid and the DG are no longer in step and the current beats with a period of about 0.07 s. It is a known risk of reclosing onto a feeder with a rotating DG.

Q17. When R2 opens, the current at node 632 does not go to zero. Why?

The plotted current is the feeder current arriving at node 632. The loads and laterals between the substation and R2 are still supplied: about 97 and 191 A in the two phases.

Q18. Without DG the fuse still blows at 1.248 s. Is that not a failure?

No, it is the intended behaviour. The fault is permanent. The fast shots give a temporary fault two chances to clear without losing the fuse. When the fault is still there, R2 goes to its delayed curve, 2.67 s at this current, and the fuse clears at 1.613 s before R2 trips again. Only the faulted lateral is lost.

Q19. How do you know the EMT result is right?

Three checks. The steady peak over RMS is $\sqrt{2}$. The pre-fault current is the load current of the load flow. And the hand check: the fuse carried fault current for $0.120 + 0.120 + 0.308 = 0.548$ s before melting, against $t_{MMT} = 0.553$ s at 2441 A from its curve.

Tricky ones**Q20. Your classification is about the DSDR. Why is the EMT run with conventional settings?**

The fault at 684 is downstream of R2. There R2 carries forward current in both schemes, and its forward group is the same, so the sequence would be identical. The EMT was made to verify the reclosing sequence and the effect of the DG, following the reference figure. An EMT run of an upstream fault with the reverse group is future work.

Q21. So does the DSDR solve the problem shown by the EMT?

No, and the report says so. For faults downstream of R2 the DG feeds the fault directly through the fuse and no setting of R2 changes that. The DSDR restores coordination for faults *between the grid and the DG*. That is why the report calls the DSDR necessary but not sufficient.

Q22. The static classification and the EMT: do they agree?

They answer different questions and they are consistent. With the zero-margin rule the fuse survives the first fast trip with DG, and the EMT also shows 68 % after one shot. The EMT adds the second shot and the dead-time infeed, which bring it to 100 %.

Q23. In a real network the DG would be tripped by its own protection. Then your result is not realistic?

The DG protection is not modelled, and I state that as a limitation. If the DG tripped during the first dead time, the dead-time infeed and the out-of-step reclosing would disappear. But the first shot would still cost 68 % instead of 21 %, because that happens before any DG protection can act. So the main finding stays: the DG uses up most of the fuse's margin.

Q24. What would you do to save the fuse in this case?

A larger or slower fuse on the lateral, checked again for selectivity with the fuses below it; a faster fast curve is not available, because R2 is already at its lowest dial; and disconnecting the DG before the first reclose. These are the fuse revision and the future work of the report.

Q25. Is the DG current during the dead time dangerous in itself?

Yes for fuse saving: a temporary fault is an arc, and the arc needs the current to stop in order to extinguish. If the DG keeps about 1.2 kA flowing, the arc may not go out, and the reclose finds the fault still there.

Q26. Is one EMT case enough?

For its purpose, yes: it verifies the sequence and shows the mechanism in time. It is not a replacement for the classification, which covers all 39 cells. More EMT cases — other locations, the reverse group, DG protection — are a natural extension.

11. Numbers to remember

Quantity	Without DG	With DG
Current through R2 (RMS)	2607 A	2436 A
R2 fast shot, curve + breaker	$0.070 + 0.050 = 0.120$ s	$0.076 + 0.050 = 0.126$ s
Fuse current at the start of the fault (RMS)	2816 A	4322 A
Fuse melting time at that current	0.361 s	0.120 s
Fuse current while R2 is open	0	about 1230 A
Fuse heat after the two fast shots	45 %	105 %
Fuse melts	1.248 s	0.752 s
Fuse has cleared	1.613 s	1.253 s
Fuse saving	works	lost

Fault at 0.30 s; dead time 0.2 s; breaker 3 cycles; step 0.1 ms; fuse F671-2 300E; R2 fast plug 300 A, TMS 0.1.